MeerFish

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This is a (non-exhaustive) description of the analysis framework used in the MeerFish code (`github.com/meerklass/MeerFish`). Please contact the author (steve.cunnington@port.ac.uk) if you feel something is missing/incorrect or needs further clarification.

We begin by defining the HI power spectrum as

$$P_{\mathrm{HI}}(k^{\mathrm{f}},\mu^{\mathrm{f}},z) = \alpha_{\parallel}^{-1}(z)\alpha_{\perp}^{-2}(z)\,\overline{T}_{\mathrm{HI}}^{2}(z)\left(b_{\mathrm{HI}}(z)+f(z)\mu^2+b_{\phi}^{\mathrm{HI}}(z)f_{\mathrm{NL}}\mathcal{M}(k,z)^{-1}\right)^2 \times P_{\mathrm{m}}(k,z)\,\mathcal{B}_{\mathrm{beam}}^{2}(k,\mu,z) \tag{1}$$

where $\overline{T}_{\mathrm{HI}}$ is the background mean HI brightness temperature, b_{HI} is the linear bias for HI, and f is the growth rate of structure. For the HI bias, we fit to hydrodynamical simulations (Villaescusa-Navarro et al., 2018) providing the model

$$b_{\mathrm{HI}}(z) = 0.842 + 0.693z - 0.0459z^2\,. \tag{2}$$

The mean HI brightness temperature is modelled using

$$\overline{T}_{\mathrm{HI}}(z) = 180\,\Omega_{\mathrm{HI}}\frac{h\,(1+z)^2}{H(z)/H_0}\,\mathrm{mK}\,, \tag{3}$$

where we use the model for Ω_{HI} based on the one used in SKA Cosmology SWG et al. (2020) but adapted based on the latest Ω_{HI} constraints at higher redshift from MeerKLASS intensity mapping (Cunnington et al., 2022)

$$\Omega_{\mathrm{HI}}(z) = 0.00067432 + 0.00039z - 0.000065z^2\,. \tag{4}$$

In Equation 1, we include the possibility for local primordial non-Gaussianity (PNG) to influence the power, which occurs by the bias acquiring a scale-dependent correction given by the $b_{\phi}f_{\mathrm{NL}}\mathcal{M}^{-1}$ term. f_{NL} modulates the PNG contribution and we assume $f_{\mathrm{NL}}=0$ as fiducial default. b_{ϕ}^{HI} is the PNG bias parameter for HI, adopted by the standard universality assumption of the halo mass function

$$b_{\phi}^{\mathrm{HI}}(z) = 2\,\delta_{\mathrm{c}}\left(b_{\mathrm{HI}}(z)-1\right)\,, \tag{5}$$

where $\delta_{\mathrm{c}}=1.686$. The final factor relating to PNG is given by

$$\mathcal{M}(k,z) = (2/3)k^2T_{\mathrm{m}}(k)D_{\mathrm{md}}(z)/(\Omega_{\mathrm{m}}H_0^2)\,, \tag{6}$$

where T_{m} is the matter transfer function and D_{md} is the linear growth factor normalized to $1/(1+z)$.

$P_{\rm m}$ (in Equation 1) is the matter power spectrum obtained from a Boltzmann solver (we used CAMB as standard Lewis et al., 2000), and the transverse fluctuations contributing to the power are affected by the telescope beam which acts as

$$\mathcal{B}_{\rm beam}(k,\mu,z) = \exp\left[-\frac{1}{2}k_\perp^2 R_{\rm b}^2\right] = \exp\left[-\frac{k^2}{2}(1-\mu^2)R_{\rm b}^2\right] \tag{7}$$

where $R_{\rm b} = \chi(z)\,\theta_{\rm FWHM}(z)/2\sqrt{2\ln 2}$. $\theta_{\rm FWHM}$ is radio telescope specific but for a generic dish size with diameter $D_{\rm dish}$ and observations at a frequency ν, we can say $\theta_{\rm FWHM} = c\,/\,(\nu D_{\rm dish})$.

Lastly, in Equation 1, the Alcock-Paczynski (Alcock and Paczynski, 1979) dilation parameters for the transverse and radial directions are given by

$$\alpha_\perp(z) = \frac{D_{\rm A}(z)/r_{\rm s}}{(D_{\rm A}(z)/r_{\rm s})_{\rm f}}\,, \quad \alpha_\parallel(z) = \frac{(H(z)r_{\rm s})_{\rm f}}{H(z)r_{\rm s}}\,. \tag{8}$$

and the true wavenumbers are distorted by

$$k_\perp = k_\perp^{\rm f}/\alpha_\perp\,, \quad k_\parallel = k_\parallel^{\rm f}/\alpha_\parallel\,. \tag{9}$$

where the fiducial indices denote the fiducial (reference) cosmology assumed in the measured wavenumbers, and the wavenumber without an f index denotes the true wavenumbers for the true cosmology. These are conventionally transposed into k and μ using the factor $F_{\rm AP} = \alpha_\parallel/\alpha_\perp$, where

$$k = \frac{k_{\rm f}}{\alpha_\perp}\left[1+(\mu_{\rm f})^2\left(F_{\rm AP}^{-2}-1\right)\right]^{1/2} \tag{10}$$

$$\mu = \frac{\mu_{\rm f}}{F_{\rm AP}}\left[1+(\mu_{\rm f})^2\left(F_{\rm AP}^{-2}-1\right)\right]^{-1/2}\,. \tag{11}$$

Finally, to correct for any change in volume caused by the different cosmologies, the power spectrum model must be multiplied by $\alpha_\parallel^{-1}\alpha_\perp^{-2}$, which explains this factor at the front of Equation 1.

Similarly to the Hi power spectrum, we can generically model the *galaxy* power spectrum as

$$P_{\rm g}(k^{\rm f},\mu^{\rm f},z) = \alpha_\parallel^{-1}(z)\,\alpha_\perp^{-2}(z)\left(b_{\rm g}(z)+f(z)\mu^2+b_\phi^{\rm g}(z)f_{\rm NL}\mathcal{M}(k,z)^{-1}\right)^2 P_{\rm m}(k,z)\,. \tag{12}$$

Observed power spectra noise and errors: Neglecting shot-noise, which will be minimal for a HI intensity mapping survey (Villaescusa-Navarro et al., 2018; Spinelli et al., 2020), the observed HI power spectrum is given as

$$P_{\rm HI}^{\rm obs}(k^{\rm f},\mu^{\rm f},z) = P_{\rm HI}(k^{\rm f},\mu^{\rm f},z) + P_{\rm N}(z)\,, \tag{13}$$

where $P_{\rm N}$ is the noise power spectrum contribution, which for HI intensity mapping is given by the telescope's thermal noise, which is modelled as

$$P_{\rm N} = V_{\rm pix}\sigma_{\rm N}^2 \tag{14}$$

where the pixel volume is determined by the pixel area $A_{\rm pix}$ so that

$$V_{\rm pix} = \Omega_{\rm A} \int_{z_0}^{z_1} {\rm d}z \frac{{\rm d}V}{{\rm d}z{\rm d}\Omega} = \Omega_{\rm A} \int_{z_0}^{z_1} {\rm d}z \frac{c\,\chi^2(z)}{H(z)}\,, \tag{15}$$

where $\Omega_{\rm A} = A_{\rm pix}(\pi/180)^2$ and the redshift intervals are determined by the frequency of observation ν and the telescope's frequency resolution $\delta\nu$ so that $z_0 = \nu_{21\rm cm}/(\nu+\delta\nu) - 1$ and $z_1 = \nu_{21\rm cm}/(\nu-\delta\nu) - 1$, with $\nu_{21\rm cm} = 1420.4\,{\rm MHz}$, the frequency of the 21cm line. The RMS of the noise fluctuations is given by the radiometer equation;

$$\sigma_{\rm N}(\nu) = \frac{T_{\rm sys}(\nu)}{\sqrt{2\,\delta\nu\,t_{\rm pix}}}\,. \tag{16}$$

The system temperature is given by

$$T_{\rm sys}(\nu) = T_{\rm rx}(\nu) + T_{\rm spl} + T_{\rm CMB} + T_{\rm gal}(\nu) \tag{17}$$

where the contribution from spill-over is $T_{\rm spl} = 3\,{\rm K}$, the background contribution from the CMB is $T_{\rm CMB} = 2.73\,{\rm K}$ and the contribution from our own Galaxy is $T_{\rm gal} = 15\,{\rm K}(408{\rm MHz}/\nu)^{2.75}$, which is tuned to the match the average sky temperature excluding $|b| < 10^\circ$ in galactic latitude. Lastly, the receiver temperature is

$$T_{\rm rx}(\nu) = 7.5\,{\rm K} + 10\,{\rm K}\left(\frac{\nu}{\rm GHz} - 0.75\right)^2\,, \tag{18}$$

which follows Cunnington (2022) which tuned this model to match recent intensity mapping observations with the MeerKAT pilot survey (Wang et al., 2021). The time per pixel $t_{\rm pix}$ in the noise RMS (Equation 16), is given by the total observation time $t_{\rm obs}$ from the survey, shared among each pixel. Thus we define it as

$$t_{\rm pix} = N_{\rm dish} t_{\rm obs}\ \left(\theta_{\rm FWHM}/3\right)^2 / A_{\rm sur} \tag{19}$$

where we assume each pixel is 1/3 the size of the telescope's beam size (approximately correct for MeerKAT map-making). In single-dish mode, we get a set of observations per dish, hence observation time per pixel is multiplied by the number of dishes $N_{\rm dish}$.

The observed counterpart for the galaxies will contain shot-noise as the noise contribution defined as

$$P_{\rm g}^{\rm obs}(k^{\rm f},\mu^{\rm f},z) = P_{\rm g}(k^{\rm f},\mu^{\rm f},z) + P_{\rm SN}(z)\,, \tag{20}$$

where $P_{\rm SN} = 1\,/\,\bar{n}_{\rm g}$ is the galaxy shot noise, with $\bar{n}_{\rm g}$ as the galaxy number density for the survey.

The errors on the power spectrum can then be estimated as

$$\sigma^2(k,\mu) = \frac{P_{\rm obs}^2(k,\mu)}{N_{\rm modes}(k,\mu)}\,, \tag{21}$$

where the number of *unique* modes is given by

$$N_{\mathrm{modes}}\left(k,\mu\right)=\frac{k^2\Delta k\Delta\mu}{8\pi^2}V_{\mathrm{sur}}\,. \tag{22}$$

where the survey volume is determined by the minimum and maximum redshift (z_{min}, z_{max}) of the survey and its area A_{sur};

$$V_{\mathrm{sur}}=\Omega_{\mathrm{sur}}\int_{z_{\mathrm{min}}}^{z_{\mathrm{max}}}\mathrm{d}z\,\frac{c\,\chi^2(z)}{H(z)}\,, \tag{23}$$

where $\Omega_{\mathrm{sur}}=A_{\mathrm{sur}}(\pi/180)^2$.

Multipole expansion: It is convenient to adopt the multipole expansion of the anisotropic power spectrum in Equation 1 (Cunnington et al., 2020; Bernal et al., 2019; Soares et al., 2021). This is achieved using

$$P(k,\mu)=\sum_{\ell}P_\ell(k)\mathcal{L}_\ell(\mu)\,, \tag{24}$$

$$P_\ell(k,z)=\frac{(2\ell+1)}{2}\int_{-1}^{1}\mathrm{d}\mu\,P(k,\mu,z)\mathcal{L}_\ell(\mu)\,, \tag{25}$$

where

$$\mathcal{L}_0=1,\quad \mathcal{L}_2=\frac{3\mu^2-1}{2},\quad \mathcal{L}_4=\frac{35\mu^4-30\mu^2+3}{8}\,. \tag{26}$$

and the errors on these multipoles is given by

$$\sigma_{P_\ell}(k)=\sqrt{\frac{(2\ell+1)}{2}^2\int_{-1}^{1}\mathrm{d}\mu\,\sigma^2(k,\mu)\mathcal{L}_\ell^2(\mu)}\,. \tag{27}$$

Fisher forecasting: To include our observable in a Fisher forecast, we consider that the set of fluctuations $\delta(\boldsymbol{x})$ within a survey map's pixels with position $\boldsymbol{x}$ can be reasonably modelled as a Gaussian distribution with mean of zero and pixel covariance $\mathbf{C}\equiv\langle\delta^*(\boldsymbol{x})\delta(\boldsymbol{x})\rangle$. The Fisher information matrix is then given by (Tegmark, 1997; Seo and Eisenstein, 2003, 2007)

$$F_{ij}=\frac{1}{2}\,\mathrm{tr}\left[\frac{\partial\mathbf{C}}{\partial\theta_i}\mathbf{C}^{-1}\frac{\partial\mathbf{C}}{\partial\theta_j}\mathbf{C}^{-1}\right] \tag{28}$$

where θ_i and θ_j are from the set of parameters $\boldsymbol{\Theta}$ that influence the covariance and their derivatives are evaluated at the fiducial values θ_{fid}, which coincides with the maximum of the likelihood distribution. The Cramer-Rao bound ensures that no unbiased method can measure θ_i with error bars less than $\sqrt{F_{ii}^{-1}}$.

For the multipole expansion formalism, we define the Fisher matrix following Taruya et al. (2011),

$$F_{ij}|_{\ell_{\mathrm{max}}}=\frac{V_{\mathrm{sur}}}{4\pi^2}\sum_{\ell,\ell'}^{\ell_{\mathrm{max}}}\int_{k_{\mathrm{min}}}^{k_{\mathrm{max}}}k^2\mathrm{d}k\,\frac{\partial P_\ell(k)}{\partial\theta_i}\left[\widetilde{C}^{\ell\ell'}(k)\right]^{-1}\frac{\partial P_{\ell'}(k)}{\partial\theta_j}\,, \tag{29}$$

where the *reduced* covariance is given by

$$\widetilde{C}^{\ell\ell'}(k) = \frac{(2\ell+1)(2\ell'+1)}{2}\int_{-1}^{1} \mathrm{d}\mu \mathcal{L}_\ell(\mu)\mathcal{L}_{\ell'}(\mu)P_{\mathrm{obs}}(k,\mu)^2\,, \tag{30}$$

For computing the derivatives for the Fisher matrix, we follow Blanchard et al. (2020) and say that given Equation 25 we have

$$\frac{\partial P_\ell(k)}{\partial\theta_i} = \frac{(2\ell+1)}{2}\int_{-1}^{1}\,\mathrm{d}\mu\frac{\partial}{\partial\theta_i}\left[P(k,\mu)\mathcal{L}_\ell(\mu)\right]\,, \tag{31}$$

since the integration limits do not depend on the parameters, and hence the derivative can be taken inside the integral. We then compute all derivatives numerically using a 5-point stencil, generically defined for a function $\mathcal{F}$ as

$$\frac{\partial\mathcal{F}}{\partial\theta} = \frac{-\mathcal{F}(\theta+2\delta\theta)+8\mathcal{F}(\theta+\delta\theta)-8\mathcal{F}(\theta-\delta\theta)+\mathcal{F}(\theta-2\delta\theta)}{12\,\delta\theta}\,, \tag{32}$$

where $\delta\theta$ is chosen such that it gives converged derivatives. Generally, we find that $\delta\theta\sim10^{-2}$ is an appropriate choice for all parameters.

Cross-correlation and multi-tracer: A cross-correlation power spectrum between the HI and galaxies is given by (dropping redshift dependence for brevity)

$$\begin{aligned}P_{\mathrm{HI,g}}(k^{\mathrm{f}},\mu^{\mathrm{f}}) = \alpha_\parallel^{-1}\alpha_\perp^{-2}\,r_{\mathrm{HI,g}}\overline{T}_{\mathrm{HI}}\left(b_{\mathrm{HI}}+f\mu^2+b_\phi^{\mathrm{HI}}f_{\mathrm{NL}}\mathcal{M}(k)^{-1}\right)\\ \times\left(b_{\mathrm{g}}+f\mu^2+b_\phi^{\mathrm{g}}f_{\mathrm{NL}}\mathcal{M}(k)^{-1}\right)P_{\mathrm{m}}(k)\,\mathcal{B}_{\mathrm{beam}}(k,\mu)\,,\end{aligned} \tag{33}$$

where $r_{\mathrm{HI,g}}(z)$ is the cross-correlation coefficient between the two tracers to modulate stochastic suppression in the cross-correlation. Similarly to the single tracer, the *multi-tracer* Fisher matrix will be given by

$$F_{ij}|_{\ell_{\max}} = \frac{V_{\mathrm{bin}}}{4\pi^2}\sum_{\ell,\ell'}^{\ell_{\max}}\int_{k_{\min}}^{k_{\max}} k^2\mathrm{d}k\frac{\partial\boldsymbol{P}_{\tau,\ell}(k)}{\partial\theta_i}\left[\mathbf{C}_{\tau\tau'}^{\ell\ell'}(k)\right]^{-1}\frac{\partial\boldsymbol{P}_{\tau',\ell'}(k)}{\partial\theta_j} \tag{34}$$

but we are now implicitly integrating over the contributions within $\tau=\{\alpha,\beta,\mathrm{X}\}$ i.e. the different different generic tracers α and β, plus their cross-correlation X. For the combination of HI intensity mapping and a galaxy survey, we would have $\alpha\equiv$ HI and $\beta\equiv$ g. The data vector $\boldsymbol{P}_{\tau,\ell}$ represents the "non-observed" power spectra (i.e. without noise) containing each tracer combination and multipole i.e.

$$\boldsymbol{P}_{\tau,\ell}(k) = \{\boldsymbol{P}_{\alpha,\ell}(k),\boldsymbol{P}_{\beta,\ell}(k),\boldsymbol{P}_{\mathrm{X},\ell}(k)\} \tag{35}$$

with each element its own $n_\ell\times n_k$ vector. For the multipoles analysis, the covariance is given by

$$\mathbf{C}_{\tau\tau'}^{\ell\ell'}(k) = \begin{pmatrix} C_{\alpha\alpha}^{\ell\ell'} & C_{\alpha\beta}^{\ell\ell'} & C_{\alpha\mathrm{X}}^{\ell\ell'}\\ & C_{\beta\beta}^{\ell\ell'} & C_{\beta\mathrm{X}}^{\ell\ell'}\\ & & C_{\mathrm{XX}}^{\ell\ell'}\end{pmatrix}\,, \tag{36}$$

where each element represents the reduced covariance as before, given by the below;

$$C^{\ell\ell'}_{\alpha\alpha}(k) = (2\ell+1)(2\ell'+1)\int_0^1 \mathrm{d}\mu\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell'}(\mu)P^{\mathrm{obs}}_\alpha(k,\mu)^2\,, \tag{37}$$

$$C^{\ell\ell'}_{\alpha\beta}(k) = (2\ell+1)(2\ell'+1)\int_0^1 \mathrm{d}\mu\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell'}(\mu)P_\alpha(k,\mu)P_\beta(k,\mu)\,, \tag{38}$$

$$C^{\ell\ell'}_{\alpha\mathrm{X}}(k) = (2\ell+1)(2\ell'+1)\int_0^1 \mathrm{d}\mu\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell'}(\mu)P^{\mathrm{obs}}_\alpha(k,\mu)P_\mathrm{X}(k,\mu)\,, \tag{39}$$

$$C^{\ell\ell'}_{\mathrm{XX}}(k) = \frac{(2\ell+1)(2\ell'+1)}{2}\int_0^1 \mathrm{d}\mu\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell'}(\mu)\left[P^2_\mathrm{X}(k,\mu)+P^{\mathrm{obs}}_\alpha(k,\mu)P^{\mathrm{obs}}_\beta(k,\mu)\right]\,. \tag{40}$$

Each element is its own sub-matrix of size $n_\ell \times n_k$. We assume no mode coupling between k, thus the only non-zero off-diagonal components in the matrix come from multipole ($\ell\ell'$) and tracer ($\tau\tau'$) covariance.

Baryon Acoustic Oscillations (BAO)

To forecast BAO detectability with intensity maps, the approach differs slightly. Following the literature (e.g. Bull et al., 2015), we generate a smoothed version of the P_ℓ power spectrum with the BAO removed. This is done by fitting a cubic spline, providing P_{smooth}. A purely oscillatory function can then be extracted by normalising by the smoothed power,

$$f_{\mathrm{BAO}}(k) = \frac{P_\ell(k) - P_{\mathrm{smooth}}(k)}{P_{\mathrm{smooth}}(k)}\,. \tag{41}$$

We plot an example of this smoothed power function in Figure 1. The detectability of the BAO can then be simply forecast with a single parameter A_{BAO}, which defines the "amplitude" of the BAO wiggles:

$$P_\ell(k) = \left[1 + A_{\mathrm{BAO}} f_{\mathrm{BAO}}(k)\right] P_{\mathrm{smooth}}(k)\,. \tag{42}$$

The Fisher forecast in Equation 29 can then be computed using this modified power spectrum with the parameter $\theta = [A_{\mathrm{BAO}}]$. Since the fiducial value for the amplitude of the BAO parameter is $A_{\mathrm{BAO}} = 1$, the signal-to-noise ratio for the BAO detectability is simply given by $\mathrm{SNR}_{\mathrm{BAO}} = 1/\sqrt{F^{-1}}$.
Note: This can be inserted into the same numerical stencil derivative as before, or the exact simple analytical solution can be used, since $\partial P_\ell\,/\,\partial A_{\mathrm{BAO}} = P_{\mathrm{smooth}} f_{\mathrm{BAO}}$. We find equivalent constraints on A_{BAO} with both numerical and analytical approaches, providing extra validation of our numerical stencil derivation.

As an extra validation step, since we focus on A_{BAO} in isolation as a single parameter, the Fisher derivation should give equivalent results to a simple summation in quadrature over the modes in f_{BAO}. i.e. $\mathrm{SNR}_{\mathrm{BAO}} = 1/\sqrt{F^{-1}}$, from

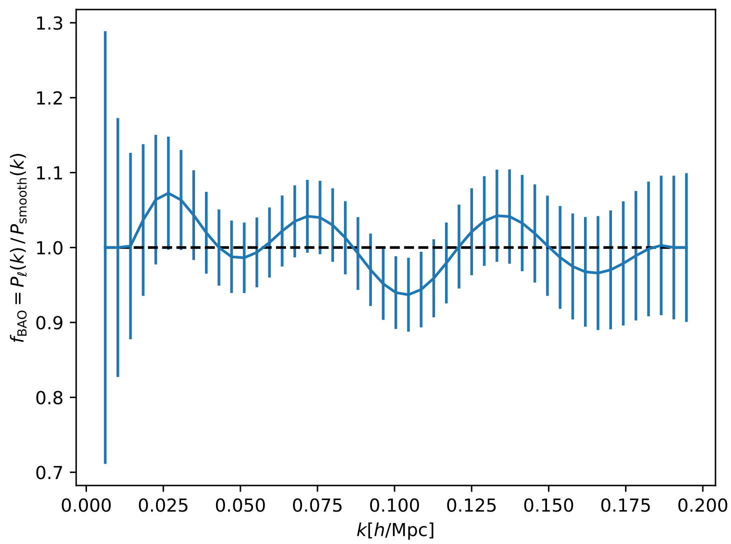

Figure 1: Example of BAO oscillatory function, $f_{\rm BAO}$, from Equation 41. Generated from this script.

the Fisher matrix, should be equivalent to

$$\mathrm{SNR}_{\rm BAO} = \sqrt{\sum_k \frac{f_{\rm BAO}(k)^2}{\sigma_{f_{\rm BAO}}(k)^2}} \equiv \sqrt{\sum_k \frac{(P_\ell(k) - P_{\rm smooth}(k))^2}{\sigma_{P_\ell}(k)^2}}\,. \tag{43}$$

We have validated that this gives consistent results with the Fisher matrix computation.

Further reading

For further reading and related works on Fisher forecasting in the context of HI intensity mapping, we recommend Camera et al. (2013); Bull et al. (2015); Fonseca et al. (2015); Alonso and Ferreira (2015); Alonso et al. (2015); Pourtsidou et al. (2017); Bernal et al. (2019).

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